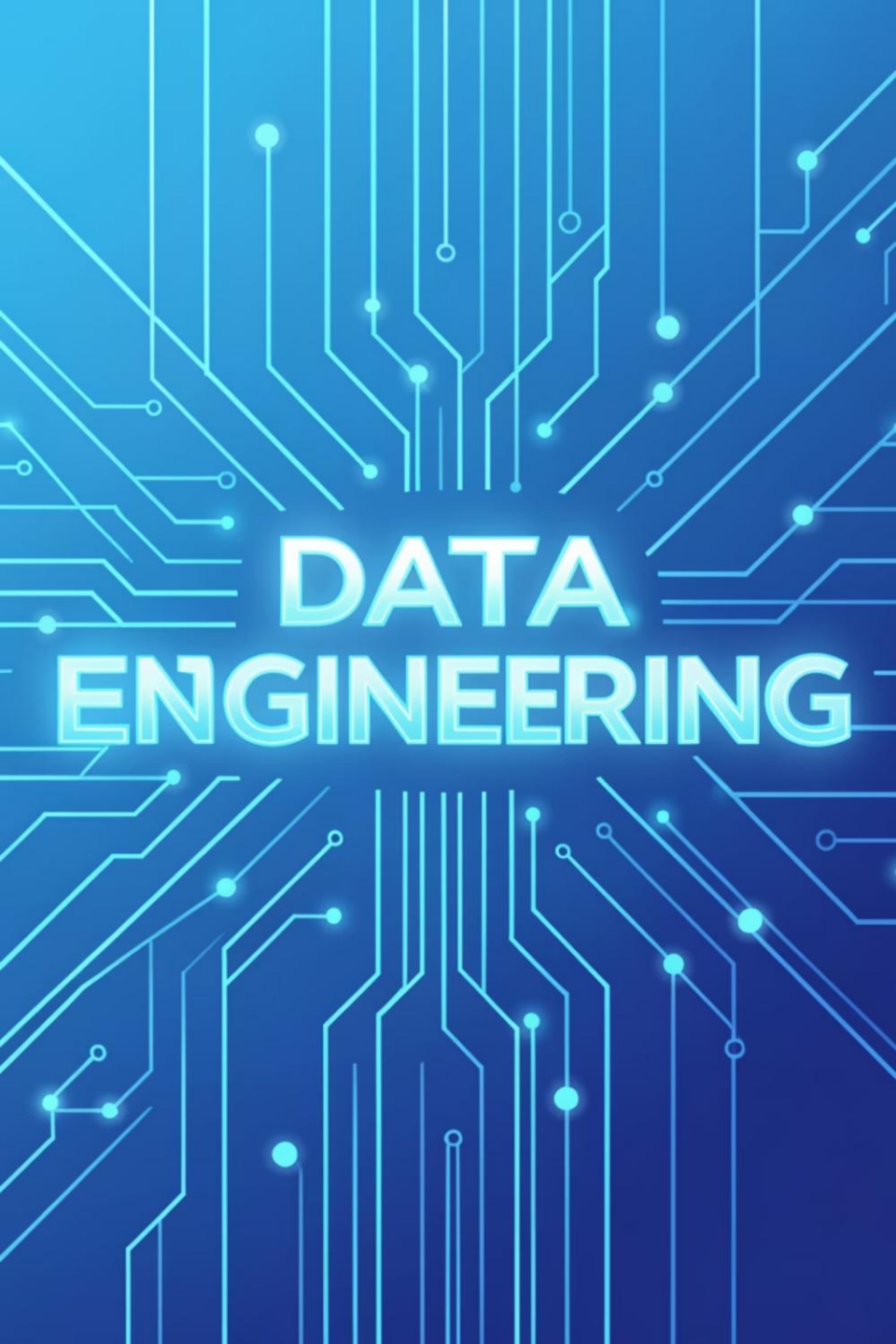




DATA ENGINEERING

Data Engineering Spring '26 – DS463

Feb 03, 2026

Lecture 7

Logistics: Your Course Essentials

Class Timings

Monday (LH6), Tuesday (LH8): 11:30 AM – 12:20 PM

Friday (LH8): 11:00 AM – 11:50 AM

Office Hours

Monday, Tuesday: 10:00 AM – 11:00 AM, S18 2nd Floor NAB or by appointment.

Credit Hours

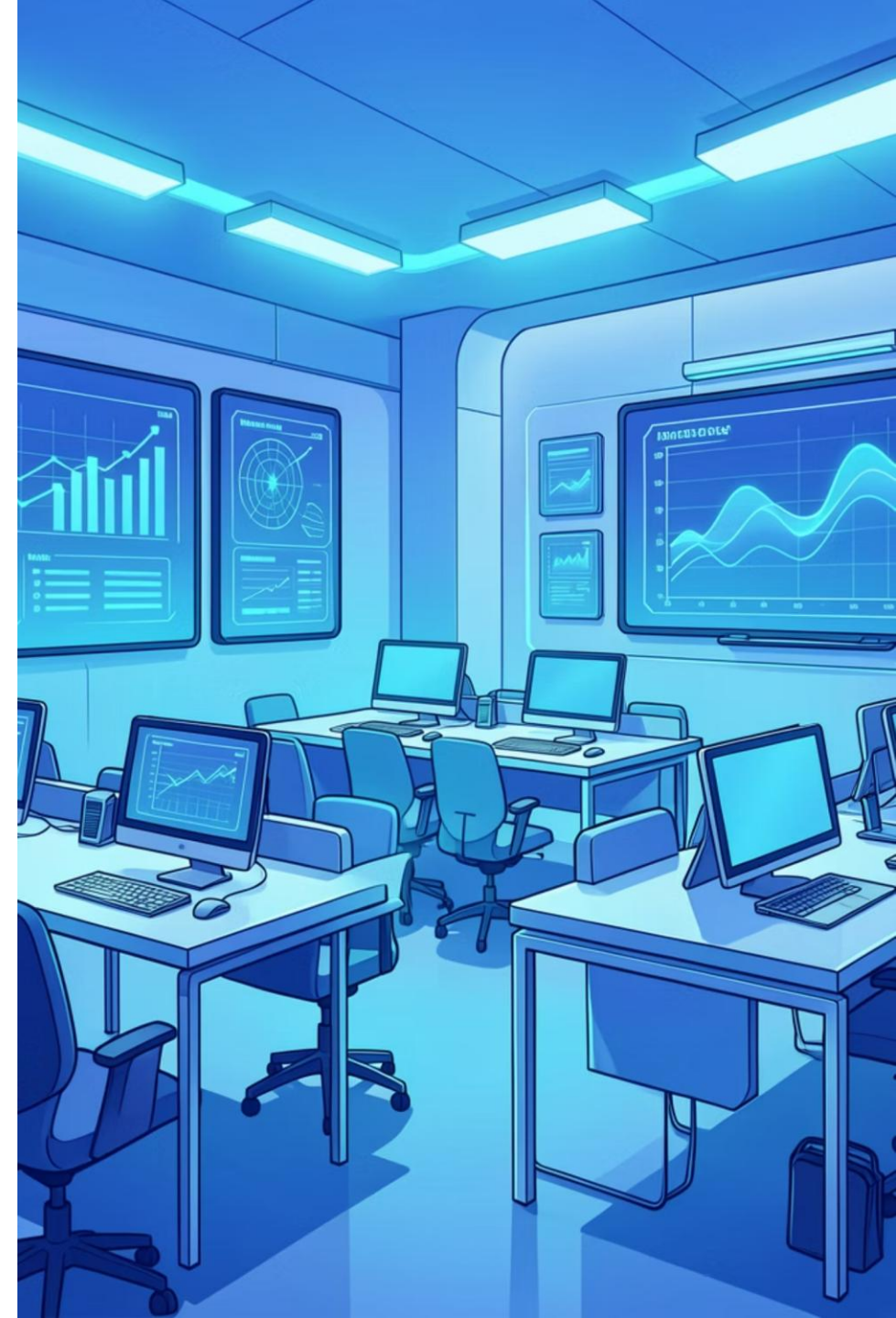
2 hours (Lectures) + 1 hour (Practical Session)

Contact & Support

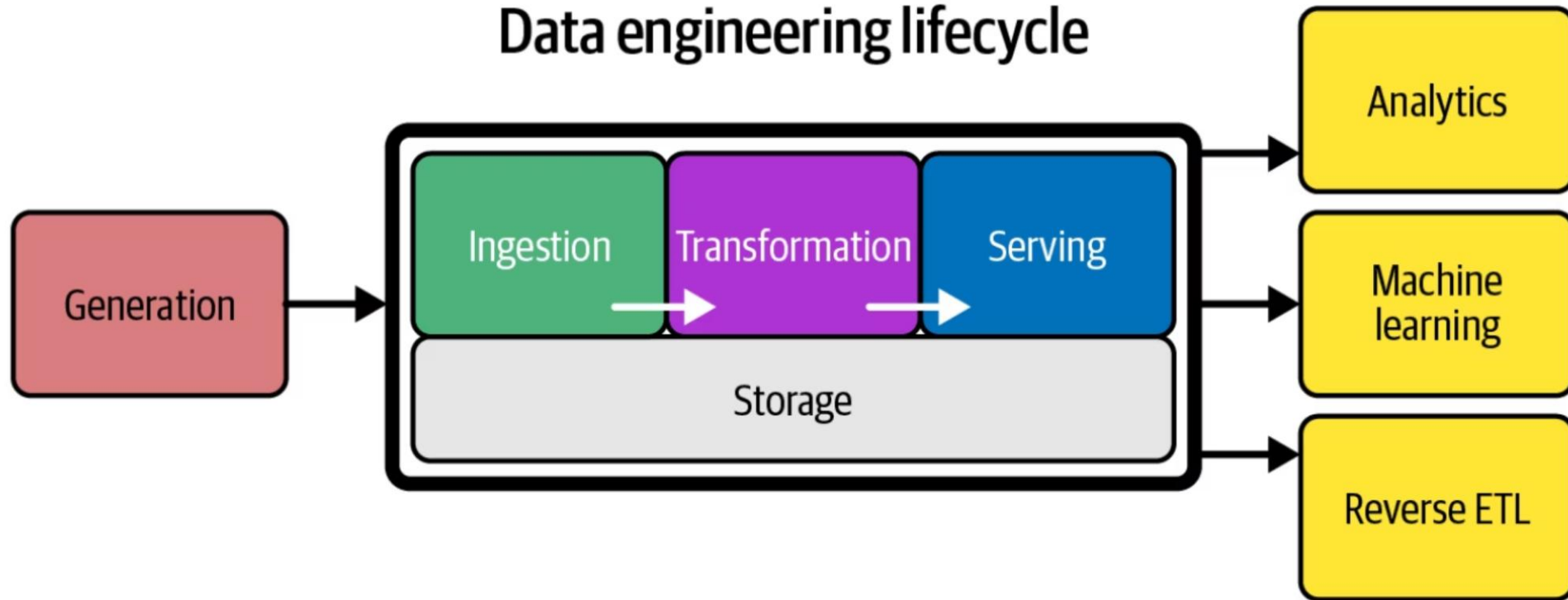
MS Teams: w5t93yk

Email: farah.saeed@giki.edu.pk

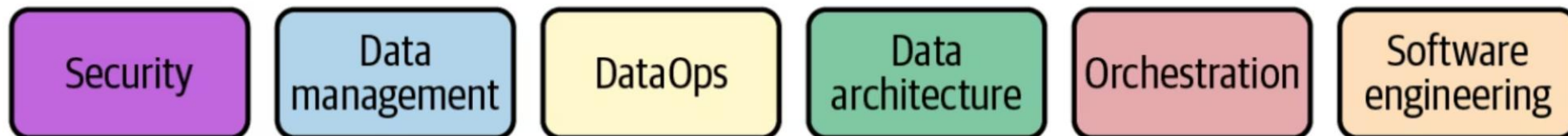
WhatsApp: Reach out for quick queries



Data engineering lifecycle



Undercurrents:



Various stages of the lifecycle may repeat themselves, occur out of order, overlap, or weave together in interesting and unexpected ways.

Recap

Source System: Data Generation:

- Iot Devices
- Apps
- Api

Storage:

- Phase continues through out the lifecycle.
- Choosing appropriate storage.
- Data access frequency.

Ingestion

- Batch Ingestion
- Streaming Ingestion
- Push and Pull Model

Transformation

What's the cost and return on investment (ROI) of the transformation? What is the associated business value?

Is the transformation as simple and self-isolated as possible?

What business rules do the transformations?

Transformation

Queries

Writing Efficient queries
SQL: Table creation, Data aggregation,
Filtering

Modeling

- Schema to denormalized schema
- Discussion with stakeholders

Transformation

- Data anomalies, ML Featurization
- Transformation before and after ingestion
- Large scale aggregation

Serving

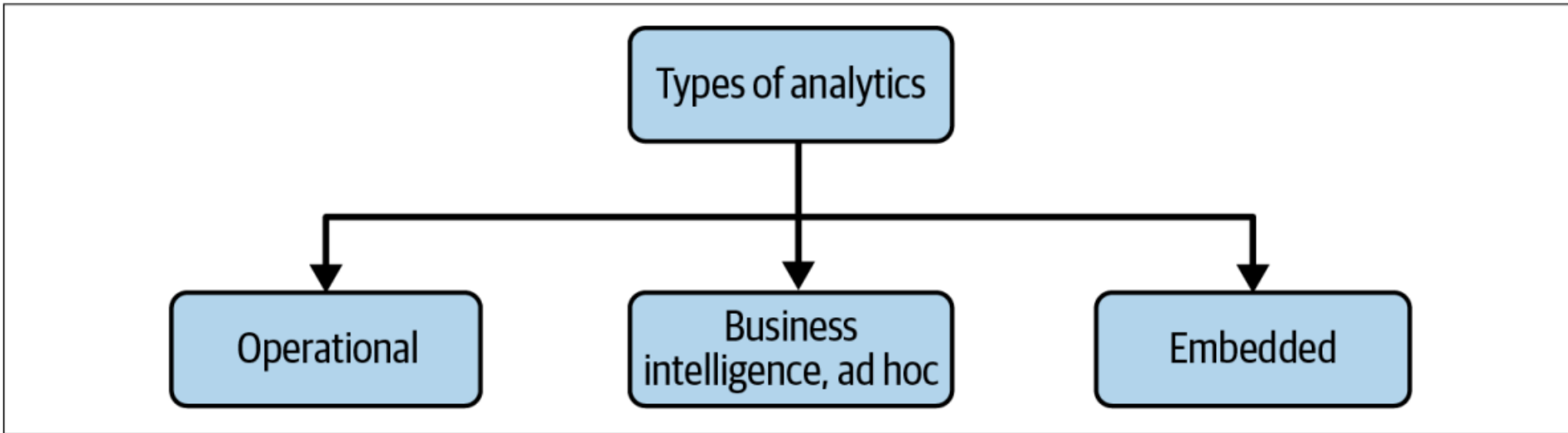


Figure 2-5. Types of analytics

Machine Learning

- Is the data of sufficient quality to perform reliable feature engineering? Quality requirements and assessments are developed in close collaboration with teams consuming the data.
- Is the data discoverable? Can data scientists and ML engineers easily find valuable data?
- Where are the technical and organizational boundaries between data engineering and ML engineering? This organizational question has significant architectural implications.
- Does the dataset properly represent ground truth? Is it unfairly biased?

Reverse ETL

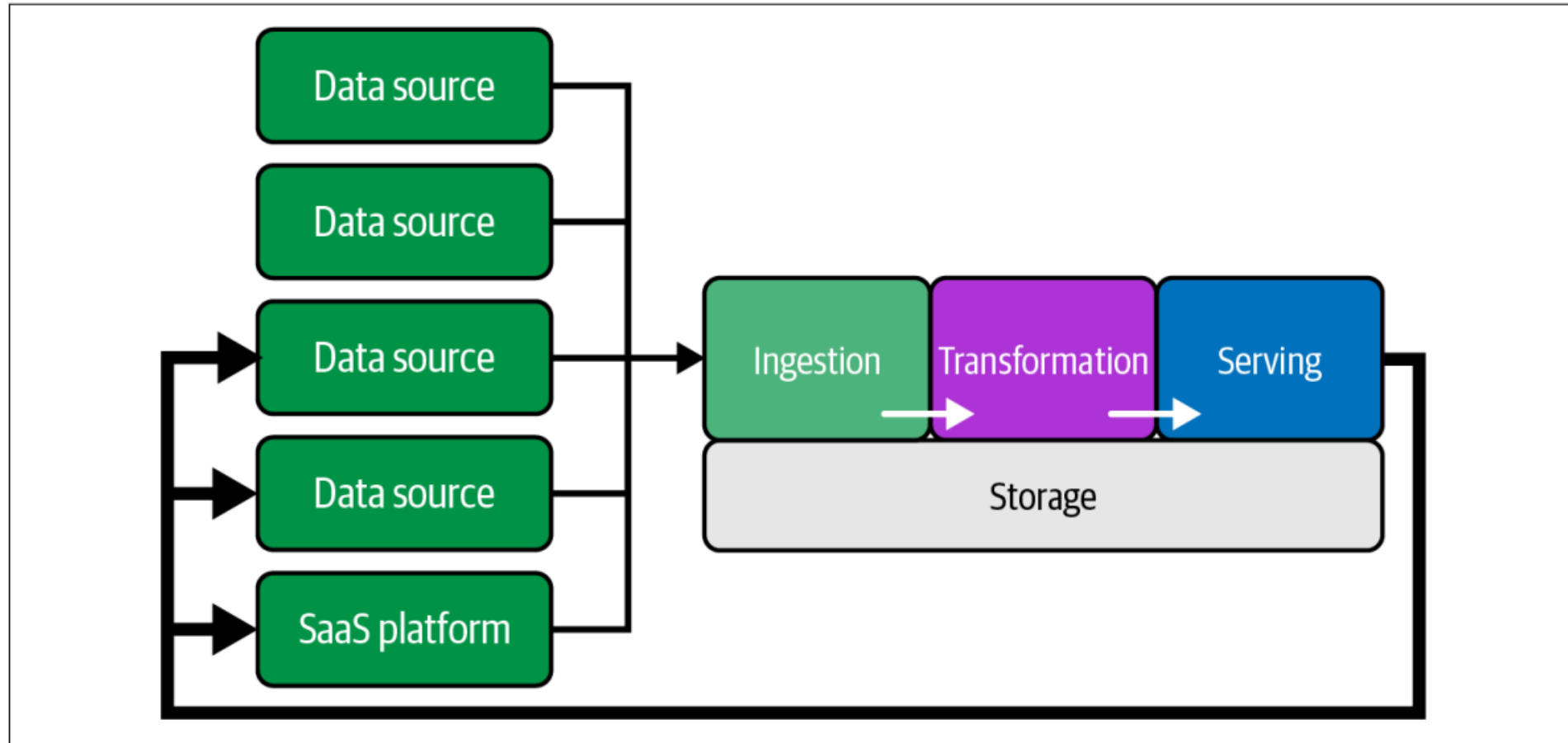


Figure 2-6. Reverse ETL

Undercurrents of Data Engineering

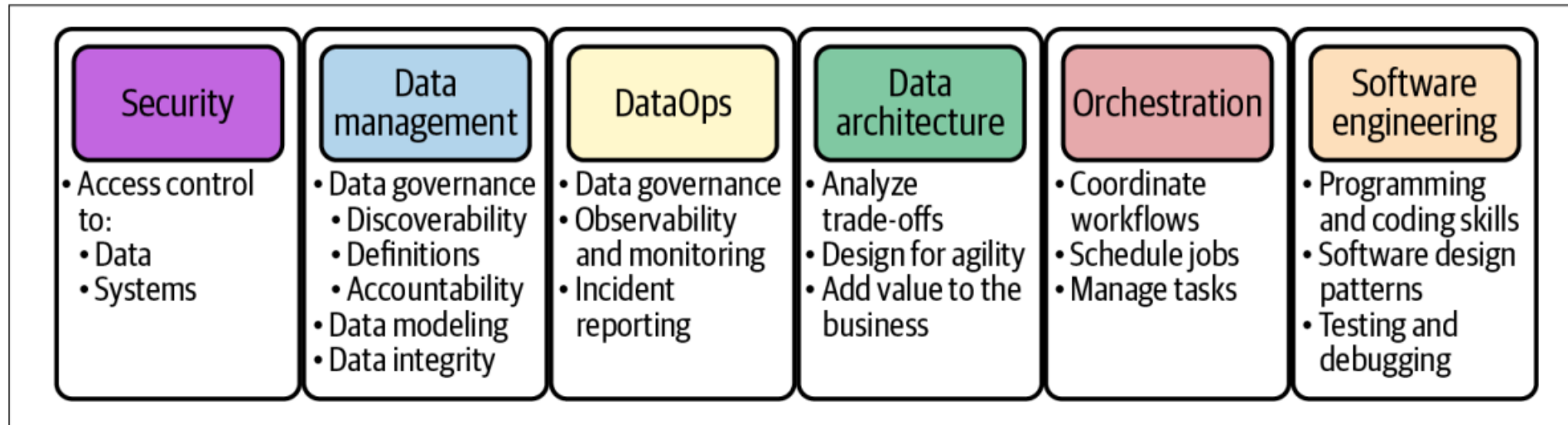


Figure 2-7. The major undercurrents of data engineering

Attempting Assessments

- Quiz
- Exams
- Answer in simplest possible way